Project task division

02805 Social Graphs and Interactions, Fall 2026 | Film collaboration networks and narrative similarity | Group of three

Research question

To what extent does the structural organization of a weighted film collaboration network, based on shared actors and directors, correspond to the lexical and semantic similarity of film plot summaries (TF-IDF and sentence-embedding cosine similarity), and do network communities reveal shared narrative themes beyond traditional genre classifications, when compared against appropriate network null models?

Key dates (from the week 8 page)

Date	Milestone
Wed 4 Nov	Dataset locked, question drafted (done). Network built: `network/films_nodes.tsv`, `films_edges.tsv`, `film_plots.zip`.
Wed 11 Nov	Project session. Shared inputs delivered (similarity matrices, weighting schemes, first partitions).
Wed 18 Nov	Project session. All analyses run, first version of the three figures, Overleaf (PNAS) set up.
Wed 25 Nov	Project session. Full draft; internal walkthrough so everyone can explain every part.
Fri 27 Nov 23:59	**Hand-in on DTU Learn:** paper (PNAS, max 6 pages incl. references), 3 slides (PDF), half-page reflection (PDF).
Wed 2 Dec	**3-minute presentation + questions.** Any member can be asked about anything.

How the work is split

The project is divided into three themes that map onto the subquestions. Each person owns one theme end to end: the code, one of the three main figures, and the matching Methods and Results text. The themes connect through a small set of shared files (below), so all three can work in parallel from day one.

	Theme	Subquestions	Main figure
A	**Network structure and null models**: weighting schemes, topology, structure vs. similarity, rewiring and genre/year nulls	SQ1, SQ2	Fig. 1 (centerpiece): plot similarity vs. network distance, with null-model bands
B	**NLP**: text cleaning, TF-IDF, sentence embeddings, LDA topics, distinctive words, hand-labelled validation	SQ1 (text side), SQ4 (topics)	Fig. 2: validation of the similarity measures + correspondence vs. director weight (with A)
C	**Clustering, communities and genre**: Louvain, stability, within/between similarity, genre alignment, themes beyond genre	SQ3, SQ4	Fig. 3: community x genre heatmap with each community's distinctive words

Shared files (the interfaces between themes)

Agree on these formats now. Row order for every matrix is `films_nodes.tsv` (sorted by `node_id`), and every file is keyed by `node_id`.

File	Owner	Used by	Due
`network/films_nodes.tsv, films_edges.tsv` (done)	A	B, C	4 Nov
`network/weightings/edges_<scheme>.tsv`: one edge list per weighting scheme	A	C	11 Nov
`network/null/rewired_<i>.tsv`: degree-preserving rewired networks	A	C	15 Nov
`nlp/sim_tfidf.npy, nlp/sim_embed.npy`: 1789 x 1789 cosine similarity	B	A, C	**11 Nov**
`nlp/topics.tsv`: LDA topic share per film; `nlp/distinctive_words.py`	B	C	18 Nov
`communities/partitions.tsv`: node_id, scheme, seed, community	C	A, B	11 Nov

The similarity matrices are the main dependency: A and C need them to start their core analyses, so B delivers a first version (TF-IDF + embeddings, no tuning) by 11 Nov and improves it afterwards.

Person A — Network structure and null models — SQ1 + SQ2

Notebook: `notebooks/02_network_analysis.ipynb` + `build_network.py`

Tasks

- **Weighting schemes (SQ2).** Extend `build_network.py` to write a set of schemes: actors only, directors only, director weight 1 / 2 / 4 / 10 relative to actors, and a billing-position variant (shared leads count more). Save each scheme's edge list for C.
- **Topology per scheme (SQ2).** Degree and strength distributions (CCDF, log-log), weighted clustering, betweenness and eigenvector centrality (top films), giant component, average shortest path (distance = 1/weight), degree assortativity. One summary table.
- **Structure vs. similarity (SQ1).** Using B's matrices: similarity of linked vs. unlinked pairs (effect size), Spearman of edge weight vs. similarity, and mean similarity by hop distance (1, 2, 3, 4+, disconnected). Mantel / QAP permutation tests, since film pairs are not independent.
- **Null models (SQ1).** (i) Degree-preserving rewiring (double-edge swaps, 500-1000 networks), z-scores of each statistic; (ii) genre- and year-matched random film pairs, to separate collaboration from genre/typecasting. Share the rewired networks with C.
- **Controlled regression.** similarity ~ linked + weight + genre overlap + |year difference|, with permutation (MRQAP) p-values.
- **Which weighting fits best (SQ2).** One pre-chosen correspondence metric (e.g. Mantel r) per scheme, plotted against director weight.
- *Optional:* disparity-filter backbone; are backbone edges more similar in plot than the removed edges?

Deliverables

- Fig. 1 (centerpiece): mean plot similarity vs. network distance, with bands for both null models.
- Network statistics table; the SQ2 panel for Fig. 2 (correspondence vs. director weight).
- Paper: **Introduction** (assembles it), Methods: network construction and null models; Results: SQ1, SQ2.

Person B — NLP: text representations, topics and validation — SQ1 text side + SQ4 topics

Notebook: `notebooks/03_nlp.ipynb`

Tasks

- **Text cleaning.** spaCy tokenization, lowercasing, stopwords, lemmatization. Remove named entities (PERSON, GPE, ORG) so character names do not create fake similarity; keep a version with names as a robustness check.

- **TF-IDF.** sklearn TfidfVectorizer (sublinear tf, min_df / max_df), cosine similarity matrix. Document the parameter choices.
- **Sentence embeddings.** sentence-transformers (e.g. all-mpnet-base-v2). Plots (~630 words) exceed the token limit: embed paragraphs and average. Cosine similarity matrix.
- **Sanity checks.** Correlation between the TF-IDF and embedding matrices; nearest neighbours of well-known films; similarity distribution within vs. across IMDb genres.
- **Hand-labelled validation (key check).** Sample ~60 film pairs stratified by similarity. All three members rate plot similarity 1-5 blind. Report inter-rater agreement and Spearman correlation of each measure with the human ratings.
- **Topics (SQ4).** LDA (gensim or sklearn), choose K by topic coherence, label the topics, save topic shares per film for C.
- **Distinctive words.** A reusable function for weighted log-odds (Monroe et al.) or class-based TF-IDF, which C applies to communities.

Deliverables

- `sim_tfidf.npy`, `sim_embed.npy` (first version by 11 Nov), `topics.tsv`, distinctive-words function.
- Fig. 2: validation panel (human ratings vs. TF-IDF and embeddings), combined with A's SQ2 panel.
- Paper: **Abstract and Significance Statement**; Methods: text processing, similarity measures, validation; Results: measure validity and topics. Compiles the half-page reflection.

Person C — Clustering: communities and genre — SQ3 + SQ4

Notebook: `notebooks/04_communities.ipynb`

Tasks

- **Community detection (SQ3).** Weighted Louvain (networkx), ~50 seeds; stability as mean pairwise NMI, or a consensus partition. Compare modularity with the modularity of A's rewired networks.
- **Semantic coherence (SQ3).** Mean within- vs. between-community similarity (TF-IDF and embeddings); permutation test that shuffles community labels while keeping community sizes. Silhouette on cosine distance as a second measure.
- **Across weighting schemes (SQ3).** Run on each of A's schemes; compare partitions with NMI/ARI and show how films move between communities.
- **Second clustering for comparison.** Cluster a k-nearest-neighbour graph built from the embeddings; NMI between plot-based clusters and collaboration communities.
- **Genre alignment (SQ4).** Genre assortativity of the network; community x genre enrichment (hypergeometric test); NMI between communities and primary genre.
- **Beyond genre (SQ4).** Compare same-genre/same-community pairs with same-genre/different-community pairs; find the most similar cross-genre pairs inside communities and read them (qualitative examples for the paper).
- **Characterize communities.** Apply B's distinctive-words function and topic shares to each community.

Deliverables

- `communities/partitions.tsv` (first version by 11 Nov).
- Fig. 3: community x genre heatmap with each community's top distinctive words; table of community statistics.
- Paper: **Discussion** (assembles it); Methods: community detection; Results: SQ3, SQ4. Assembles the 3 slides.

Shared responsibilities

Task	Who	Notes
Literature	All	3-4 real references each in your own area: collaboration networks (A), text similarity / embeddings (B), community detection and homophily (C).
Paper assembly	A: Introduction B: Abstract, Significance C: Discussion	Each person writes the Methods and Results for their own theme. PNAS template on Overleaf, max 6 pages incl. references.
Hand labels	All	Everyone rates the same ~60 pairs blind (organized by B).
Code review	A reviews B B reviews C C reviews A	One review each by 18 Nov, so a second person understands every analysis.
Slides	C assembles	3 plain slides, one per main figure; each person drafts the slide for their figure.
Reflection	B compiles	Half page on the "From an AI-drafted report" boxes; each adds 2-3 sentences.
Walkthrough	All, 25 Nov	Each person presents their part to the others; anyone can be asked about anything on 2 Dec.

Timeline

Week	Person A (network)	Person B (NLP)	Person C (clustering)
4-11 Nov	Weighting schemes; topology table; rewiring code	Text cleaning; TF-IDF and embedding matrices (**first version due 11 Nov**)	Louvain pipeline and stability; genre assortativity baseline
11-18 Nov	SQ1 analyses with both nulls; regression; SQ2 correspondence	Hand-labelled validation; LDA topics; distinctive words	Within/between similarity; schemes comparison; beyond-genre analysis
18-25 Nov	Fig. 1 final; write Intro, Methods, Results	Fig. 2 final (with A); write Abstract, Significance, Methods, Results	Fig. 3 final; write Discussion, Methods, Results
25-27 Nov	Polish; references	Reflection; final read-through	Slides; final read-through

Checks per grading category

Category	How we cover it	Lead
Research question	Specific RQ with four subquestions; restated in the Introduction and answered in the Discussion	A
Data and scope	Own dataset combining Wikidata, IMDb and Wikipedia; network + text; 1,789 films	All
Showcasing the course	Null models, centrality, communities, backbone (network); TF-IDF, embeddings, topics, NER (NLP)	All
Analysis and checks	Degree-preserving and genre/year nulls (A); hand-labelled sample + two text measures (B); seed stability + second clustering (C)	All
Figures	Three figures, each answers part of the question; labelled axes, self-explanatory captions	Owner of each figure
References	Real, relevant, correctly cited, placing the work in context	All
Overall impression	One story: does who-works-with-whom predict what the film is about?	C (Discussion)
Presentation	3 slides; everyone can explain every part (walkthrough 25 Nov)	C

Draft author contributions (to adapt at hand-in)

The course does not accept "all members contributed equally". A starting point:

A built the collaboration network and weighting schemes, carried out the network-structure and null-model analyses (Fig. 1) and wrote the Introduction. B developed the text-processing pipeline, the TF-IDF and sentence-embedding similarity measures, the topic model and the hand-labelled validation (Fig. 2), and wrote the Abstract and Significance Statement. C performed the community detection, the genre and theme analyses (Fig. 3) and wrote the Discussion. All authors labelled the validation sample, wrote the Methods and Results for their analyses, and revised the paper.